

Fluctuation Spectra of Membranes

Our HH was

$$f_c = 2k_c(H - C_0)^2 + k_g K$$

where f_c is the curvature free energy per unit area.

Here we are going to use f_c to calculate the fluctuation spectrum $\langle |h_q|^2 \rangle$ for a given temperature.

Simplifications for this case:

- $C_0 = 0$

- 1D membrane, so $K = 0$

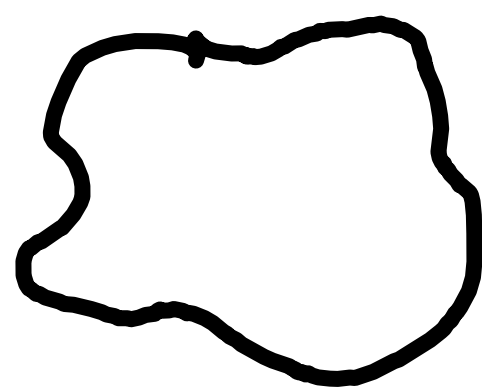
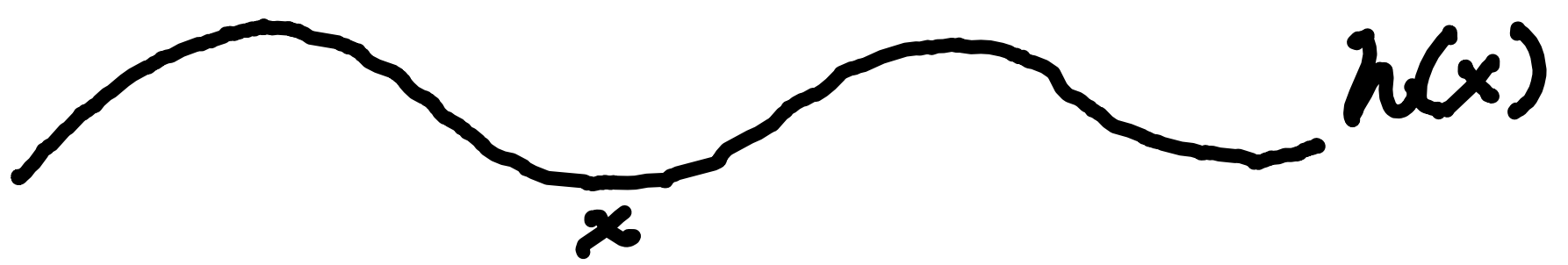
- mostly flat:

so

$$H = \frac{h_{xx}}{2}$$

not

"Monge gauge"



So, with these simplifications:

$$f_c = \frac{k_c}{2} (h_{xx})^2$$

and the total F.E. is

$$F = \int_0^L dx f_c = \frac{k_c}{2} \int_0^L dx (h_{xx})^2$$

how do we deal with this? its even messier in 2D.

inverse Fourier transform to the rescue!!!

$$FT: h(q) = \frac{1}{\sqrt{L}} \int_0^L dx h(x) e^{iqx}$$

$$IFT: h(x) = \frac{1}{\sqrt{L}} \sum_q h(q) e^{-iqx}$$

Using the IFT,

$$\begin{aligned} h_x &= \frac{1}{\sqrt{L}} \sum_q h(q) \frac{d}{dx} e^{-iqx} \\ &= \frac{-i}{\sqrt{L}} \sum_q h(q) q e^{-iqx} \end{aligned}$$

$$h_{xx} = \frac{-1}{\sqrt{L}} \sum_q h(q) q^2 e^{-iqx}$$

benefit: we could evaluate the 2nd derivative! Without even knowing what $h(x)$ is!!!

COOL

So, we plug h_{xx} back in and get:

$$E_c = \frac{k_c}{2L} \int_0^L dx \left[\sum_q q^2 h(q) e^{-iqx} \right]^2$$

aka

$$= \frac{k_c}{2L} \int_0^L dx \sum_q \sum_{q'} q^2 q'^2 h(q) h(q') e^{-i(q+q')x}$$

aka

$$= \frac{k_c}{2L} \sum_q \sum_{q'} q^2 q'^2 \underbrace{\int_0^L dx e^{-i(q+q')x}}_{\text{Dirac delta function}}$$

Dirac delta function

L if $q = -q'$

0 for $q \neq -q'$

So, all terms in which $q \neq -q'$ are 0, and

$$F_c = \frac{k_c}{2} \sum_q q^4 \underbrace{h(q)h(-q)}_{\substack{h(q)h^*(q) \\ \text{aka} \\ |h(q)|^2}}$$

$$F_c = \frac{k_c}{2} \sum_q q^4 |h(q)|^2$$

free energy in Fourier space!

remember - longer wavelength $\rightarrow$ lower q
so q^4 increases quickly for higher
frequency modes.

Now there's a rule in stat mech / thermal physics called the

"equipartition theorem":

- every "quadratic" degree of freedom contributes, on average, the same amount to F at equilibrium

- the amount it contributes is $k_B T / 2$

$h(q)$ is a quadratic d.o.f.! and each term contributes $k_B T / 2$:

$$F = \frac{k_c}{2} \sum_i q_i^4 / h(q_i)^2$$

$$\text{so, } \frac{k_c}{2} q_1^4 \langle 1/h(q_1)^2 \rangle = \frac{k_c}{2} q_2^4 \langle 1/h(q_2)^2 \rangle$$

$$= \frac{k_c}{2} q_3^4 \langle 1/h(q_3)^2 \rangle$$

$$= \frac{k_c}{2} q_4^4 \langle 1/h(q_4)^2 \rangle$$

and so on

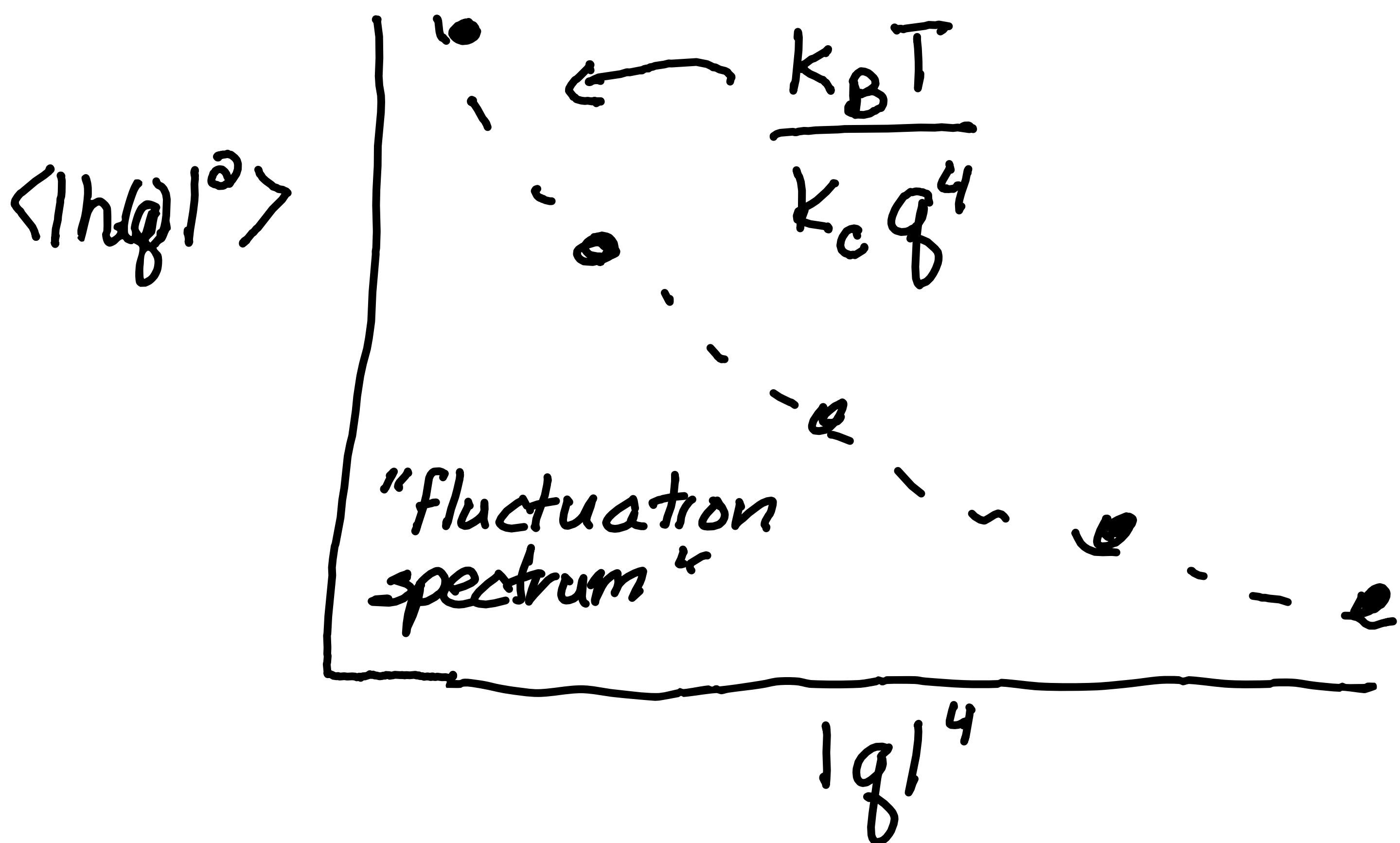
$$= k_B T / 2.$$

That is, for any q_i :

$$\frac{k_c}{2} q_i^4 \langle 1/h(q_i)^2 \rangle = \frac{k_B T}{2}$$

$$\langle |h(q_i)|^2 \rangle = \frac{k_B T}{k_c q_i^4}$$

we can calculate this from $h(x)$ data over many frames



The fit coefficient is $\frac{k_B T}{k_c}$!!!

2D

Most of the time we are interested in 2D membranes.

2D FT:

$$\text{FT: } h(q_x, q_y) = \frac{1}{\sqrt{A}} \int dx \int dy h(x, y) e^{-iq_x x} e^{-iq_y y}$$

$$\text{IFT: } h(x, y) = \frac{1}{\sqrt{A}} \sum_{q_x} \sum_{q_y} h(q_x, q_y) e^{iq_x x} e^{iq_y y}$$

Using the IFT,

$$\lambda_{xx} = -\frac{1}{\sqrt{A}} \sum_{q_x} \sum_{q_y} \lambda(q_x, q_y) q_x^2 e^{iq_x x} e^{iq_y y}$$

$$\lambda_{yy} = -\frac{1}{\sqrt{A}} \sum_{q_x} \sum_{q_y} \lambda(q_x, q_y) q_y^2 e^{iq_x x} e^{iq_y y}$$

So, in the Monge gauge

$$H = \frac{\lambda_{xx} + \lambda_{yy}}{2} = \frac{1}{2\sqrt{A}} \sum_{q_x} \sum_{q_y} \lambda(q_x, q_y) (q_x^2 + q_y^2) e^{iq_x x} e^{iq_y y}$$

which can be written like so:

$$H = \frac{1}{2\sqrt{A}} \sum_{\vec{q}} \lambda(\vec{q}) |\vec{q}|^2 e^{i\vec{q} \cdot \vec{r}} \quad \text{where } \vec{r} = x, y \text{ and } \vec{q} = q_x, q_y$$

Then

$$F_c = \frac{k_c}{2\sqrt{A}} \int dA \sum_{\vec{q}} \sum_{\vec{q}'} \lambda(\vec{q}) \lambda(\vec{q}') |\vec{q}|^2 |\vec{q}'|^2 e^{i(\vec{q} + \vec{q}') \cdot \vec{r}}$$

Now, again

$$\int dA e^{i(\vec{q} + \vec{q}') \cdot \vec{r}} = \sqrt{A} \delta_{\vec{q}, -\vec{q}'}$$

$$F_c = \frac{k_c}{2} \sum_{\vec{q}} |\lambda(\vec{q})|^2 |\vec{q}|^4$$

and just as in 1D

$$\langle |\lambda(\vec{q})|^2 \rangle = \frac{k_B T}{k_c \vec{q}^4}$$

$$\text{where } \vec{q}^4 = (\vec{q} \cdot \vec{q})^2 = [q_x^2 + q_y^2]^2$$